

Number (version)	75CLR00674_Att1 (3.0)
Title	Next Generation Pack 2 (NGP2) and Secondary Packaging Machine 2 (SP2) General Overview

1.0 Purpose

The purpose of this overview is to provide information on overview aspects of the processes being carried out by the Next Generation Pack Machine 2 (NGP2) / Secondary Packaging Machine 2 (SP2) and the products being used and manufactured by them.

2.0 Scope

This is an information document and applies to the processes carried out by the Next Generation Pack Machine 2 / Secondary Packaging Machine 2 (SP2) and the products / raw materials being used by their processes.

3.0 Definitions

- NGP2 – Next Generation Pack Machine 2
- SP2 – Secondary Packaging Machine 2
- WIP – Work in Progress
- 70% IPA – 70% Isopropyl Alcohol

4.0 References

75CLR00674

5.0 Roles and Responsibilities

Production representatives for the Next Generation Pack Machine 2 / Secondary Packaging Machine 2 (SP2) ensures that the information in this document is current.

	EMERGENCY STOP In the event of an emergency push this button to STOP the machine.
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Number (version)	75CLR00674_Att1 (3.0)
Title	Next Generation Pack 2 (NGP2) and Secondary Packaging Machine 2 (SP2) General Overview

Next Generation Pack Machine 2 (NGP2)

This machine is a 3-person operation.

The operator(s) load sleeved catheters onto the vibratory tracks, catheters are placed on the belt and in position, they are fed through the following processes of adding foam hydration solution and then packaging, the top and bottom film will be applied, and the packages will be sealed and cut, finger hole punched out and labels will be applied to the pack and sent out to SP2

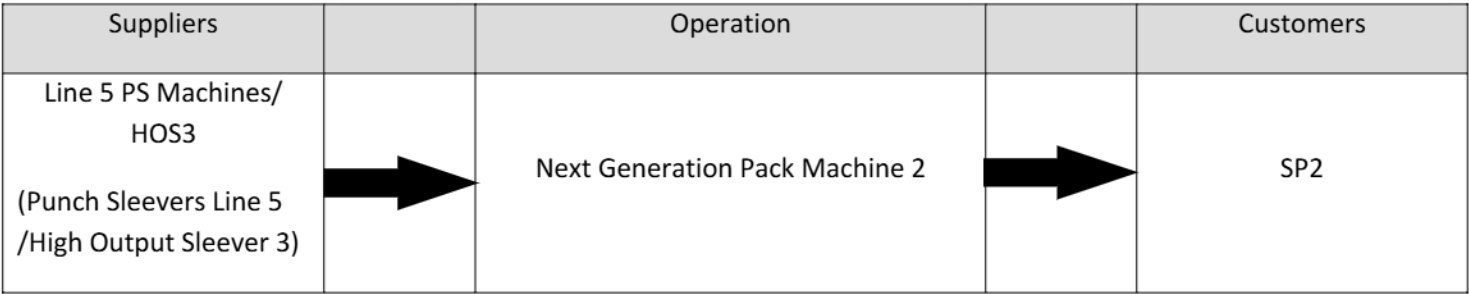
Prior to the packaging being sealed the foil will go through the printers to apply the fixed and variable print and the branding label.

Once the catheters are sealed and packaged, they will be sent out to SP2 to be placed in boxes. The in-process quality checks will be performed in this area, at regular intervals through-out the shift as per the document 75CLR00674_Att5,6,7 (Low, medium, high risk)



Number (version)	75CLR00674_Att1 (3.0)
Title	Next Generation Pack 2 (NGP2) and Secondary Packaging Machine 2 (SP2) General Overview

INTRODUCTION



Learning Objectives:

At the end of this training activity the Trainee will be able to:

- Operate the Next Generation Pack Machine in conjunction with the Work order in a safe and efficient manner.
- Operate the machine to produce a minimum target set by the Team Leader

Responsibilities:

Team Leader and Training Coordinators:	Manufacturing Instruction Implementation.
Operators:	Adherence to the Manufacturing Instruction.

Training Method:	Equipment:
Training Method will involve: • Job Talk • Practical Demonstration • Stamina Development	Listed below are the equipment requirements necessary for this operation: • Next Generation Pack Machine/ Secondary packaging Machine • Gloves • Safety Glasses • Isopropanol 70% • Lint free wipes • Scissors

Number (version)	75CLR00674_Att1 (3.0)
Title	Next Generation Pack 2 (NGP2) and Secondary Packaging Machine 2 (SP2) General Overview

MATERIALS FOR NEXT GENERATION PACK MACHINE 2 (NGP2)

Catheters

Are coated with LCC on VCL5. Catheters used on VCL5 are different to catheters used in other machines as they are non-PVC catheters with non-PVC funnels more commonly known as TPE. The catheters then move to PS5/PS6/PS7/PS8/HOS3 to be sleeved and a tip and cap applied.

Ensure to scan WIP ticket into machine and SAP to ensure correct catheters are used.

Each tote box has a WIP Ticket attached which contains the relevant details i.e., part number, machine number, Lot number, the date, shift and the quantity made.

Catheter Subs are in different diameters and lengths.

CHARRIER SIZE	DESCRIPTION	FUNNEL COLOUR
CH08-40 91A	VaPro SWT	BLUE
CH10-40 83A	VaPro SWT	BLACK
CH12-40 83A	VaPro SWT	WHITE
CH14-40 83A	VaPro SWT	GREEN
CH16-40 83A	VaPro SWT	ORANGE
CH08-40 91A	F-STYLE	BLUE
CH10-40 88A	F-STYLE	BLACK
CH12-40 88A	F-STYLE	WHITE
CH14-40 88A	F-STYLE	GREEN
CH16-40 88A	F-STYLE	ORANGE
CH08-20 91A	VaPro SWT	BLUE
CH10-20 83A	VaPro SWT	BLACK
CH12-20 83A	VaPro SWT	WHITE
CH14-20 83A	VaPro SWT	GREEN

Number (version)	75CLR00674_Att1 (3.0)
Title	Next Generation Pack 2 (NGP2) and Secondary Packaging Machine 2 (SP2) General Overview

MATERIALS FOR NEXT GENERATION PACK MACHINE 2 (NGP2)

Hydration Solution (H2)

H2 Hydration Solution – is stored in the Warehouse in 250L Grey IBC tanks and is transported to the Next Generation Packaging Machine via the material pass for use at NGP2.

Hollister H2 Hydration solution consists of predominately water with small amounts of surfactant & other ingredients to enhance hydration of the coating.

Check Hydration Solution 2 label is present on the drum before use.

Ensure the label contains the expiry date and the date is within expiry before use.

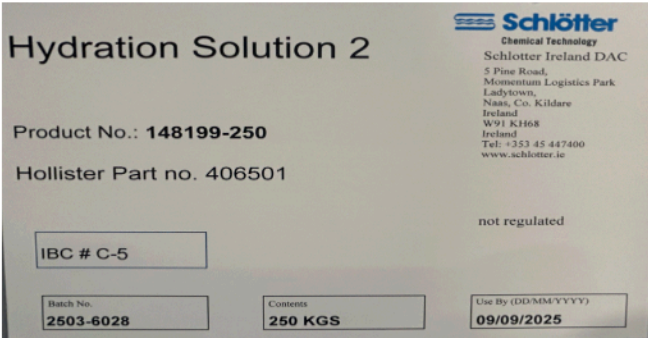
Note: expiry date will be stamped or written by R&D on the drum.

Film Roll

The same rolls of foil are used for both male and female packs.

The roll of film is fed through the machine using various rollers that are labelled with numbers and directional arrows.

Remember to scan the MRP number against the current production work order.



Number (version)	75CLR00674_Att1 (3.0)
Title	Next Generation Pack 2 (NGP2) and Secondary Packaging Machine 2 (SP2) General Overview

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Labels

Brand labels are applied to the packaging. The rolls of labels must be checked throughout the shift to avoid running out. The HMI will display a warning when the level of labels at the label station are running low or empty. This label contains the information for the customer that it is Hollister product and the catheter that is inside.

Remember to scan the MRP number against the current production work order.

Print Ribbon

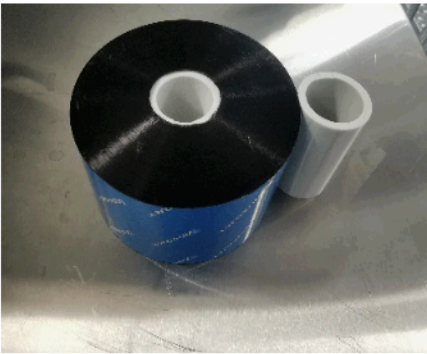
There are 7 Print ribbons in total that are used to apply the fixed and variable print, the registration mark, and the arrow on the foil. When ribbon is running low the status will be displayed on the HMI. If the ribbon runs out the machine will stop immediately, and an alarm message will be raised on the HMI. For the fixed and variable print, there are 2 stations. The left-hand printer applies the print to lanes 2&4, and the right-hand printer applied the print to lanes 1,3 & 5. Station 35 variable print (55mm). Station 37 fixed print (110mm). Station 32 print mark (22mm). Remember to scan the MRP number against the current production work order.

Foam Generator/ Foam Module

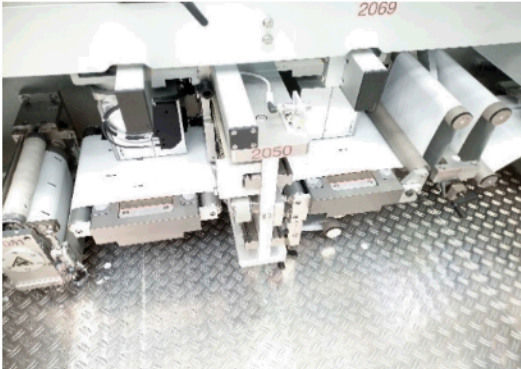
The foam module contains all components necessary to generate, deliver, dispense, and evacuate foam. If it is being changed out to be cleaned it can be wheeled away. It is supplied with a 250L IBC of hydration fluid which draws hydration solution from the IBC with a solution lance and transferred into the foam generator.



Brand labels



Ribbon for NGP2



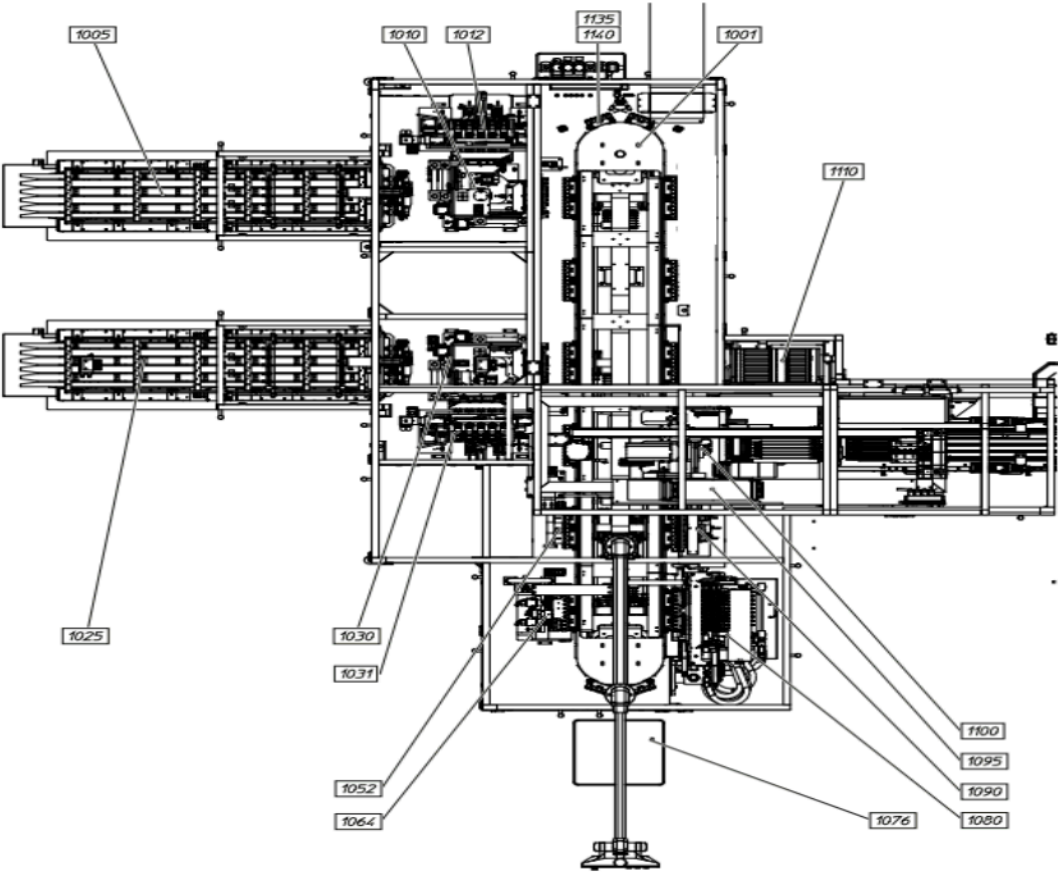
Printers on NGP2



Foam cart

Number (version)	75CLR00674_Att1 (3.0)
Title	Next Generation Pack 2 (NGP2) and Secondary Packaging Machine 2 (SP2) General Overview

Machine overview



Number (version)	75CLR00674_Att1 (3.0)
Title	Next Generation Pack 2 (NGP2) and Secondary Packaging Machine 2 (SP2) General Overview

The machine consists of the following stations:

Station 1001	Main drive	Station 2068	Waste grid detection
Station 1005	Feeding catheter rear	Station 2069	Outfeed belt 1
Station 1010	Load Catheter rear	Station 2070	Pack presence check outfeed
Station 1012	Align catheter and cap rear	Station 2070	Pack Presence Check Outfeed
Station 1025	Feeding catheter front	Station 2071	Pack Presence Check Before Reject
Station 1030	Load catheter front	Station 2073	Pack reject 1
Station 1031	Align catheter and cap front	Station 2075	Pack presence check 1
Station 1032	Align Catheter and Cap Rear / Front	Station 2077	Outfeed belt 2
Station 1052	Positioning tip	Station 2081	Pack reject 2
Station 1064	Remove cap and inflate sleeve	Station 2083	Pack presence check 2
Station 1076	Foam Generator	Station 2085	Outfeed belt 3
Station 1080	Foam dosing	Station 2087	Slug detection
Station 1090	Re-assemble cap	Station 2105	Web guiding top foil
Station 1095	Check cap presence	Station 2107	Web guiding top foil
Station 1100	Pick and place catheter	Station 2109	Web guiding bottom foil
Station 1110	IPC and hand load station	Station 2111	Web guiding bottom foil
Station 1135	Check format position	Station 2113	Web guiding top foil
Station 1140	Check catheter cap holder empty	Station 2115	Top foil web guiding
Station 2020	Loading tray	Station 2152	Top foil unwind
Station 2022	Catheter pusher	Station 2154	Bottom foil unwind
Station 2026	Splice detection top foil	Station 2170	Rewind
Station 2029	Top foil pull rolls	Station 2180	Encoder
Station 2032	Print mark printing		
Station 2035	Fixed printing		
Station 2037	Variable printing		
Station 2039	Top foil pull rollers		
Station 2041	Vision system printing		
Station 2042	Top label presence check		
Station 2043	Waste grid pull rolls		
Station 2045	Brand label application		
Station 2047	Sticky tab application		
Station 2048	Vision check arrow		
Station 2049	Label presence check		
Station 2050	Arrow printing		
Station 2051	Bottom foil pull rollers		
Station 2053	Thumb Hole Punching		
Station 2055	Bottom foil pull rollers		
Station 2056	Splice detection bottom foil		
Station 2057	Rotary sealing		
Station 2059	Pack pull rollers		
Station 2060	Detection Sealing Area		
Station 2061	Pack cooling		
Station 2062	Detection of filled pouches		
Station 2063	Pack pull rollers		
Station 2065	Pack advance		
Station 2067	Pack contour cut		

Number (version)	75CLR00674_Att1 (3.0)
Title	Next Generation Pack 2 (NGP2) and Secondary Packaging Machine 2 (SP2) General Overview

MACHINE OVERVIEW

Catheter Feeding Rear (St.1005) Load catheter rear (ST.1010) and Catheter Feeding Front (1025)

Catheters are loaded into the machine via two infeed stations with 5 individual vibratory tracks per station.

The catheters are transported to the end of the tracks. The station has two positions and take five catheters each, the catheters are collected by top of the grippers, the station then rotates 90 degrees and transfers the catheters to the alignment station, then onto the oval runner, sensors check whether the catheters are present in the grippers.

Sensors on the tracks detect when catheter levels are low on the feed tracks. When the low level is reached a warning is raised on the machine HMI.

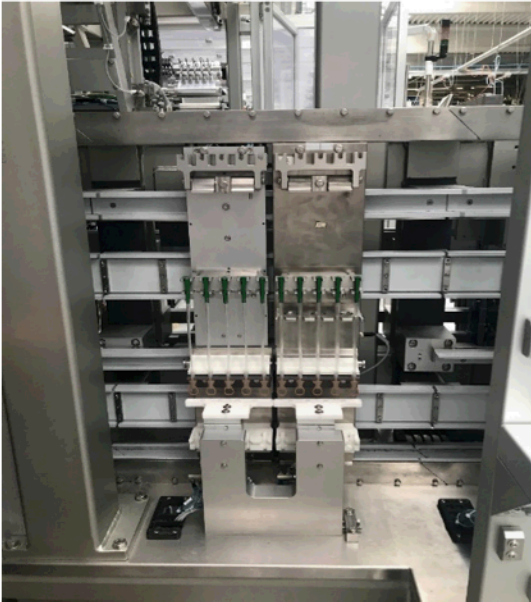
If catheters are not added to the tracks the machine will alarm and stop for continual low level on the catheter feed track.

MAIN DRIVE CARRIERS

When the catheters reach the end of the vibratory infeed tracks, they are place onto carriers, the carrier takes 5 catheters at a time from each infeed station, so ten catheters in total are placed onto the carrier.

The cap is removed for insertion of foam and once the foam has been inserted into the tip of each of the catheters the cap is replaced.

If the operator(s) notices any leaks or spills of foam hydration it must be cleaned immediately from the machine, or if the machine stops and the foamed catheters are on the carrier for any longer than 30 minutes they will be rejected.



Number (version)	75CLR00674_Att1 (3.0)
Title	Next Generation Pack 2 (NGP2) and Secondary Packaging Machine 2 (SP2) General Overview

**Align Catheter and Cap Rear (St. 1012) Front (St. 1032),
Remove cap and inflate sleeve (St.1064)**

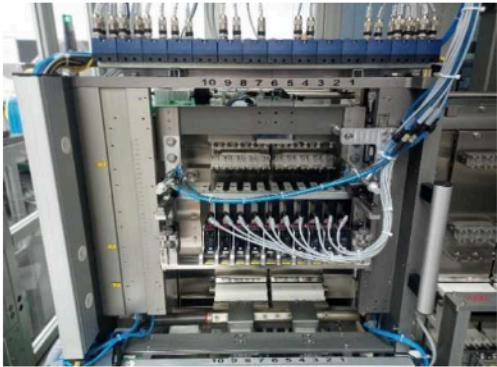
Cameras look at the cap which signal the servo drive to align. Air is blown into the sleeve whilst grippers hold the catheter in position. Caps are removed, gripped, and turned into position. Caps which are not turned correctly are pulled off and dropped into a reject chute.

Foam Dosing (St. 1080)

The catheters are filled with foam in this station. The foam is generated by two modules. The catheters are filled from below. Sensors detect the filling level separately in each lane. The excess foam is conveyed back again. The dosing module can be swivelled for maintenance. The catheters are held in place with grippers while the foam is being dosed into the tip.

Dosing Nozzle Plates

As part of the design of the fluid paths, there are changeable dosing nozzle plates (based on catheter CH size) at the end of the path that the sleeved catheters will sit into before the foam is dispensed. There are three different sizing for the dosing nozzle plates, with there being two of each type being used when in production. There are Ch08-Ch10, Ch12-14 and a Ch16 dosing nozzle plate size options for use. The dosing Nozzle Plates are cleaned per 75CLR00674_Att9 prior to loading onto the machine and after removal from the machine i.e change-over.



Red – Inlet/Outlet
Green – Flat Surface (Top)



Yellow – Groove Points
Blue – Connection Points
Green – Flat Surface (Bottom)

Number (version)	75CLR00674_Att1 (3.0)
Title	Next Generation Pack 2 (NGP2) and Secondary Packaging Machine 2 (SP2) General Overview

MACHINE OVERVIEW

Re assemble Cap (St.1090) and check cap presence (St.1095)

Caps that had been previously removed are now placed back onto the tip of the foamed catheter before the catheters are moved on to be sealed. Sensors check that they are put back in place successfully.

Pick and Place Catheter (St. 1100)

In this station catheters are transported from the hydration module to the loading tray (St. 2020). Bad parts are recognised and dropped into the discharge chute to be rejected.

IPC and Hand Load Station (St. 1110)

Filled foamed catheters are placed on a 10-catheter tray for quality testing. The request for catheters can be accessed by pressing a button outside the guard, the whole process can be accessed without interrupting or stopping the machine.

Check Format Position (St. 1135)

This checks the position of the funnel holders, to see if the holders for the catheters have been folded up or down as per the format.

Check Cap Holder Empty (St. 1140)

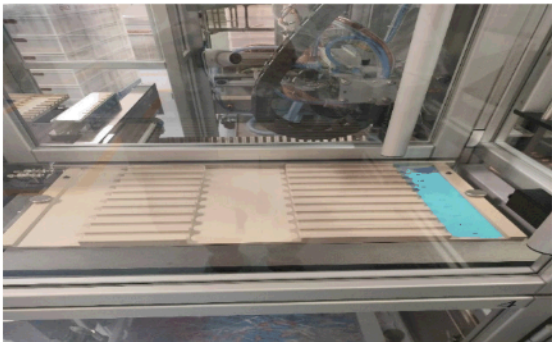
This station checks that the cap holder is empty prior to loading new catheters onto the carrier.

Loading tray (St. 2020)

Catheters are inserted using the pick and place station. Five catheters at a time are pushed to the foil path, then the next 5 catheters are pushed out to the foil path. Sensors detect the position of the plate.

Funnel Pusher (St. 2022)

Catheters are gently pushed into the sealing rollers at different times, the lower parts of the guide pushes catheters 1,3 and 5 and the upper part pushes 2 and 4. The guide pushes catheters until they are directly in front of the sealing station.



Number (version)	75CLR00674_Att1 (3.0)
Title	Next Generation Pack 2 (NGP2) and Secondary Packaging Machine 2 (SP2) General Overview

MACHINE OVERVIEW

Rotary Sealing (St.2057)

This station is a rotating format sealing station that welds together two foils in non-stop operation. The station consists of 2 replaceable format sealing cylinders that are each mounted on a heated drum. The cylinders turn in opposite direction to the foil. The complete station is designed as an interchange station to facilitate a quick format change.

This is a high temperature region on the machine and the sealing cylinders must not be contacted unless you are using the correct PPE

Splice Detection Top Foil (St.2026) and Bottom Foil (St.2056)

Sensors detect a splice on the top and bottom of the foil web. Products near the splices are marked as bad parts in the shift register and are individually ejected at a later stage, if there are consecutive errors the machine will stop.

Top Foil Pull Rolls (St.2029)

The station is used to transport the top machine web into the sealing station. The speed of the material is monitored by the rotary encoder and adjusted as necessary.

Top Foil Pull Rollers (St.2029) and Bottom Foil Pull Rollers (St.2051) (St.2055)

The station is used to ensure transport of the top and bottom film into the next station, the speed of the material is monitored by the print mark and adjusted if necessary.

Vision System Printing (St.2041)

During each machine index 5 packs are produced. On each pack variable and fixed product information is printed, the camera inspects the variable and fixed printed applied to the packs and confirms the information is correct and legible. Variable print contains: BATCH number, Expiry Date, Production Date, Data Matrix Code, GTIN. Fixed print contains product handling and manufacturing site information.



Number (version)	75CLR00674_Att1 (3.0)
Title	Next Generation Pack 2 (NGP2) and Secondary Packaging Machine 2 (SP2) General Overview

MACHINE OVERVIEW

Variable Printing (St.2037)

Two thermos transfer printers are mounted perpendicular to the direction of pouch film flow. They print variable data on the pouch material. Two printers are required due to the offset from lane 1, 3, 5 to lane 2 and 4.

The first printer prints the information on lanes 1,3 and 5 and the second printer prints the information on lanes 2 and 4.

Fixed Printing (St.2035)

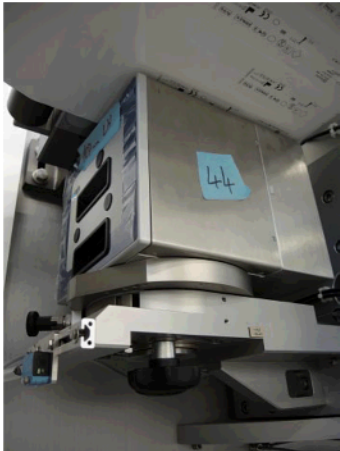
Two thermos transfer printers are mounted perpendicular to the direction of pouch film flow. They print fixed data on the pouch material. Two printers are required due to the offset from lane 1, 3, 5 to lane 2 and 4.

The first printer prints the information on lanes 1, 3 and 5 and the right-hand printer prints the information on lanes 2 and 4, during printing the foil is secured by means of a vacuum on a print table.

Print Mark Printing (St.2032)

The thermal transfer printer adds print marks to the pouch material. The print marks are printed continuously on the pouch material.

The print mark is used to control the position of the web as it is transported through the machine.



Number (version)	75CLR00674_Att1 (3.0)
Title	Next Generation Pack 2 (NGP2) and Secondary Packaging Machine 2 (SP2) General Overview

MACHINE OVERVIEW

Web Guiding Top Foil (St.2105) and (St.2107)

THE WEB EDGE CONTROLLER IS USED TO CONTROL THE PATH OF THE TOP PACKAGING MATERIAL FOIL TO THE TARGET POSITION IN THE FOLLOWING STATION – **2035 – FIXED PRINTING.**

THE WEB EDGE CONTROLLER (ST.1107) IS USED TO CONTROL THE PATH OF THE TOP PACKAGING MATERIAL FOIL TO THE TARGET POSITION IN THE FOLLOWING STATION – **2029 – TOP FOIL PULL ROLLS**

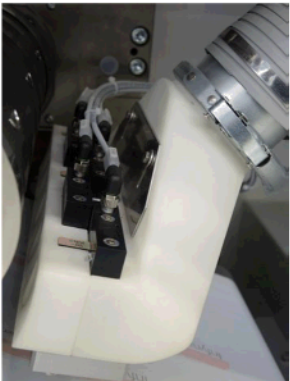
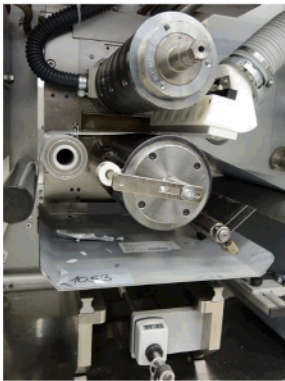
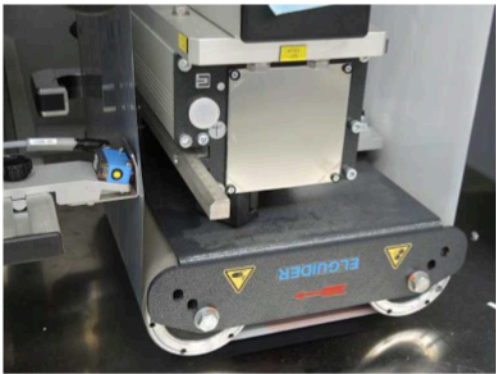
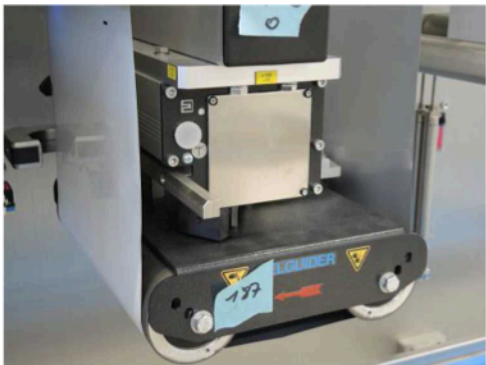
WEB GUIDING BOTTOM FOIL (ST.2109) AND (ST.2111)

THE WEB EDGE CONTROLLER (ST.1109) IS USED TO CONTROL THE PATH OF THE BOTTOM FOIL PACKAGING MATERIAL TO THE TARGET POSITION IN THE FOLLOWING STATION – **2045 – BRAND LABEL APPLICATION.**

THE WEB EDGE CONTROLLER (ST.2111) IS USED TO RECTIFY THE PATH OF THE BOTTOM FOIL TO THE TOP TO THE TARGET POSITION IN THE FOLLOWING STATION – **2053 – THUMB HOLE PUNCHING**

Bottom Foil Unwind (St.2154) and Top Foil Unwind (St.2152)

In this station the foil is continually unwound from one of the two unwinding shafts. The station consists of two pneumatic easy glide expansion shafts that are driven via servo motors, an automatic splicing device and a magazine.



Number (version)	75CLR00674_Att1 (3.0)
Title	Next Generation Pack 2 (NGP2) and Secondary Packaging Machine 2 (SP2) General Overview

MACHINE OVERVIEW

Brand Label Application (St.2045)

Two labelling units are mounted perpendicular to the direction of pouch film flow. They affix the brand label on the pouch material. Two printers are required due to the offset from lane 1, 3, 5 to lane 2 and 4. One unit dispenses on lane 1,3, 5 and the other unit dispenses on lanes 2 and 4. During labelling the foil is secured by means of a vacuum on a print table. The backing material is extracted by a suction pipe and shredded.

Label Presence Check (St.2049)

The presence of the brand labels is checked by measuring sensors. A roller lever is triggered for each position on the web when a label is present on the bottom web.

Arrow printing (St. 2050) and vision check arrow (St.2048)

Two thermal printers print arrows on the foil, each individual pack will have an arrow indicating the direction of the pack to be opened by the user.
The data on the pack is checked for validity by three cameras simultaneously.



Brand label application



Arrows printed on the foil

Number (version)	75CLR00674_Att1 (3.0)
Title	Next Generation Pack 2 (NGP2) and Secondary Packaging Machine 2 (SP2) General Overview

MACHINE OVERVIEW

Pack Pull Rollers (St.2063)

Located on the outfeed from the sealing station. The station consists of a pair of rollers. The upper roller is cooled, while the bottom roller has rubber inserts. The rollers only contact the side seals on the sealed web in between the individual pack seals. There is no contact on the 2 outer liner seal positions.

The station is closed while the machine is running and provides a constant tension to pull the web from the sealing station.

Detection sealing area (St.2060)

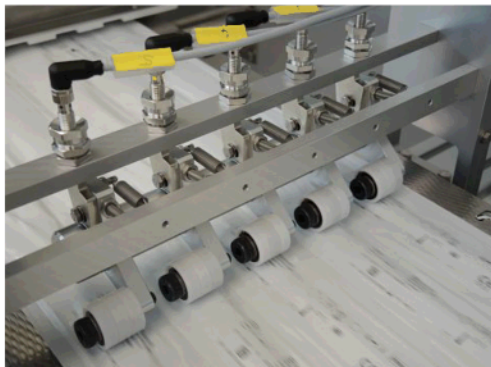
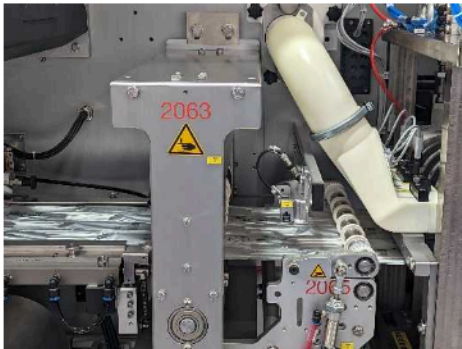
In this station rollers are used to check whether the catheters are in lateral sealing, a roller runs over the web in each lane. If there is a catheter present the roller is raised, and a sensor transmits a signal.

Pack Cooling (St.2061)

In this station, the sealed pouch foil is cooled with an air flow and active cooling plate. The upper cooling unit can be moved upwards by hand. Cooling liquids flow through the cooling plate.

CATHETER PRESENCE CHECK (ST.2062)

The presence of the products at the infeed of the 2062 station is indicated by sensors, one for each lane. This station consists of rollers that detects the presence of catheters. Station 62 individually checks for the presence of catheters in each of the machine lanes using sensors connected to rollers. If a catheter is not present the roller will not lift to trigger the sensor and the pack is rejected automatically at the product reject station.



Number (version)	75CLR00674_Att1 (3.0)
Title	Next Generation Pack 2 (NGP2) and Secondary Packaging Machine 2 (SP2) General Overview

MACHINE OVERVIEW

Pack Advance (St.2065)

In this station the sealed packs are fed into the cutting station.

Pack contour cut (ST.2067)

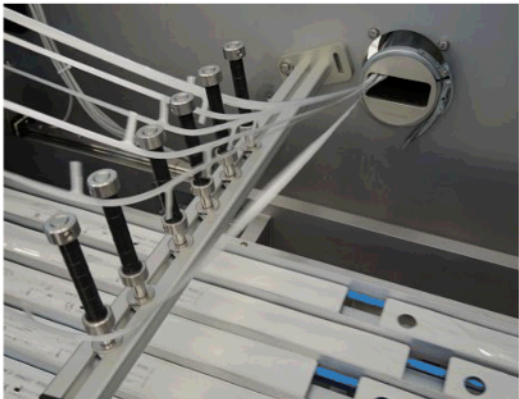
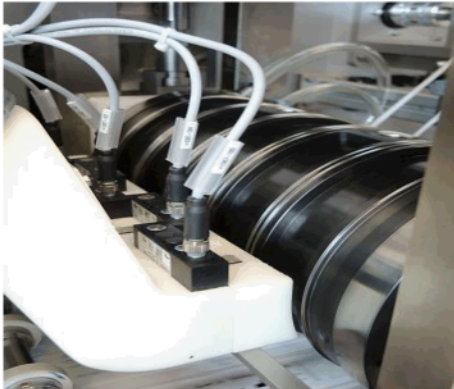
The packs are cut out of the foil at this station, two punch holes are cut out of the foil, the position of the punch holes are controlled by the print mark.

Slug Detection (St.2087)

The waste from **St.2067 – Pack Contour Cut** is removed by the vacuum extractor. One nozzle for each web. The waste sensor is mounted at the nozzle. It checks whether the waste from station 2067 passes through the nozzle of the extractor.

WASTE GRID DETECTION (ST.2068)

THE WASTE GRID FOIL IS CHECKED FOR TEARS/BREAKAGES WITH THE AID OF ROTATING DRUMS AND SENSORS. THE WASTE GRID IS FED OVER ROLLERS, WHEN THE MACHINE IS RUNNING THE WASTE GRID CREATES ROTATIONAL MOVEMENT OF THE ROLLERS. SENSORS DETECT WHETHER THE ROLLS ARE MOVING BY MEANS OF PERMANENT SIGNAL CHANGE. IF NO MOVEMENT IS DETECTED THE MACHINE WILL ALARM AND STOP FOR WASTE GRID FAULT. THE WASTE GRID IS EXTRACTED VIA A VACUUM SUCTION.



Number (version)	75CLR00674_Att1 (3.0)
Title	Next Generation Pack 2 (NGP2) and Secondary Packaging Machine 2 (SP2) General Overview

MACHINE OVERVIEW

Outfeed Belt 1 (St.2069) Outfeed Belt 2 (St.2077)

This station is situated between the pack contour cut and pack reject. The discharge belt receives the separated pouches from the contour cutting station and the pouches are segregated into five lanes.

Pack Presence Check outfeed (St.2070) and Pack Presence Check Before Reject (St.2071)

The presence of the products at the infeed of station 2069 is checked with sensors, one for each lane.

The station checks the presence of products before **St.2073 – Pack Reject**

Pack Reject (St.2073) and Pack Presence Check in Reject (St.2075)

If a bad part is detected by the machine, the product is marked as a reject in machine register and rejected at station 2073. The station can individually reject a pack from any machine lane. The reject products make their way out of the machine via a belt.

In the event of a power cycle please ensure to manually remove all packs after St.2073

IF A PRODUCT IS NOT DETECTED WHEN EXPECTED BY THE SENSORS LOCATED IN ST 2075, THE MACHINE WILL STOP AND ALARM FOR A MISSING REJECT PACK



Number (version)	75CLR00674_Att1 (3.0)
Title	Next Generation Pack 2 (NGP2) and Secondary Packaging Machine 2 (SP2) General Overview

MACHINE OVERVIEW

IPC (St.2081) and Pack Presence Check in IPC (2083)

On request, a set number of products can be requested for sampling. When a product is requested, the product is ejected with aid of gates; one for each lane. The products are sent down a slide into a collection hopper. The station (1083) checks the presence of products that pass via the slide of St.1081 – IPC by means of sensors.

Outfeed Belt (St.2085)

The belt conveyer transports the products out of the machine and into SP2.



Number (version)	75CLR00674_Att1 (3.0)
Title	Next Generation Pack 2 (NGP2) and Secondary Packaging Machine 2 (SP2) General Overview

HMI OVERVIEW

NGP2 has 2 HMI screens, one at the manual infeed station and one located in the middle of the machine which can move as needed.

Each HMI can control the whole machine.

- 1) **Emergency Stop Button** – This button stops the entire machine in the event of danger. The entire machine stops immediately.
- 2) **Reset Button** – Button acknowledges information or error messages, by pressing the reset button error messages are acknowledged.
- 3) **Start Button** – Button starts the machine and flashes in the start and stop phase.
- 4) **Stop Button** – When pressed the machine will stop and remain in the home position.
- 5) **Touchscreen** - The Panel View and HMI is used for operating and monitoring the machine.



Number (version)	75CLR00674_Att1 (3.0)
Title	Next Generation Pack 2 (NGP2) and Secondary Packaging Machine 2 (SP2) General Overview

HMI OVERVIEW

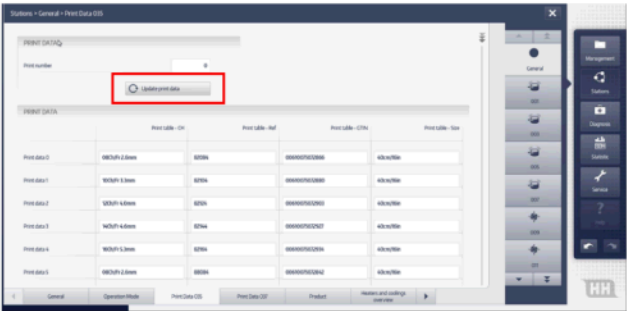
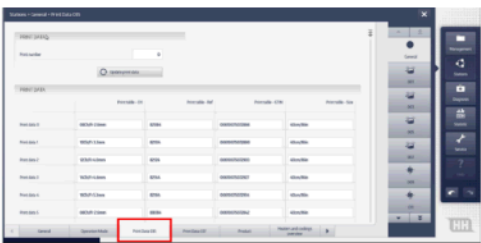
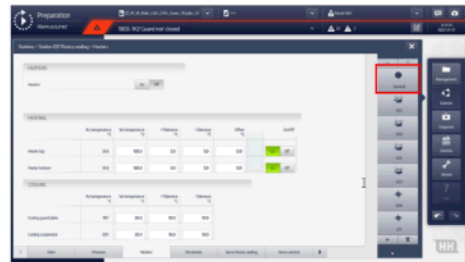
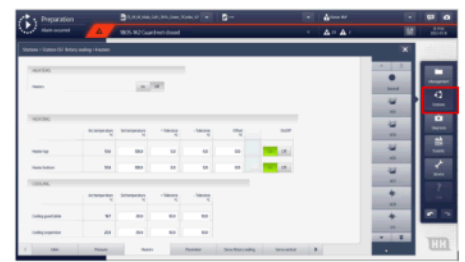
Updating Lot Number

The lot/number update needs to be completed at 7:59am.
If the operators do not complete the print update the machine will not start. An alarm will appear on the HMI stating that the print data needs to be updated.

1. Press the stations button on the HMI (Figure 1)
2. Press the general button (Figure 2)
3. Press the Print Data 035 Tab on the bottom of the screen of the HMI (Figure 3)

Updating Lot Number

1. Press the update data button (Figure 4)
2. Print data is now updated to the correct manufacturing date
3. Confirm print data on the packaging

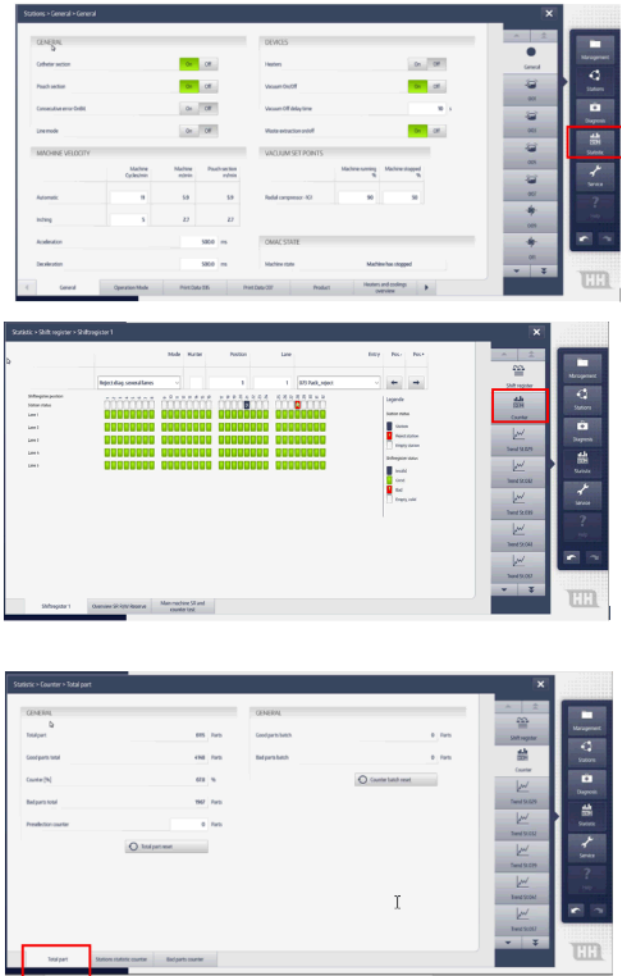


Number (version)	75CLR00674_Att1 (3.0)
Title	Next Generation Pack 2 (NGP2) and Secondary Packaging Machine 2 (SP2) General Overview

HMI OVERVIEW

Checking Counter

- 1. Press the statistic button on the HMI (Figure 5)
- 2. Press the counter button (Figure 6)
- 3. Total parts, Good parts and Bad parts will be displayed on this page. (Figure 7)



Number (version)	75CLR00674_Att1 (3.0)
Title	Next Generation Pack 2 (NGP2) and Secondary Packaging Machine 2 (SP2) General Overview

MATERIALS FOR SECONDARY PACKAGING MACHINE

Cartons

The catheters will be packed into the cartons. The 2 different types of cartons include:

- Vapro (20cm)
- Vapro (40cm)

Cartons are brought onsite as flat pack, loaded into the machine by the operator, picked and glued into shape by the machine.

Literature

There are 2 types of Literature for SP2

- Vapro IFU which is placed into Lane 1
- Generic PCN which is placed into Lane 2

Place small stacks on the conveyer sloping backwards to aid the machine when picking them up individually.

Push the stacker up against the last stack to ensure the machine feeds it in correctly and efficiently.

Glue Granules (Technomelt)

Ensure the glue unit is powered on, the power switch is on the bottom right-hand side of the unit
Normal temperature of the unit is 150 degrees and requires heat proof gloves to be worn.

The glue is in the form of granules and with heat it will turn to a hot melt adhesive.
The glue granules will be added manually into the glue unit as seen in the picture, by the operator.



Literature



Number (version)	75CLR00674_Att1 (3.0)
Title	Next Generation Pack 2 (NGP2) and Secondary Packaging Machine 2 (SP2) General Overview

MACHINE OVERVIEW

Carton Building (F3 Robot)

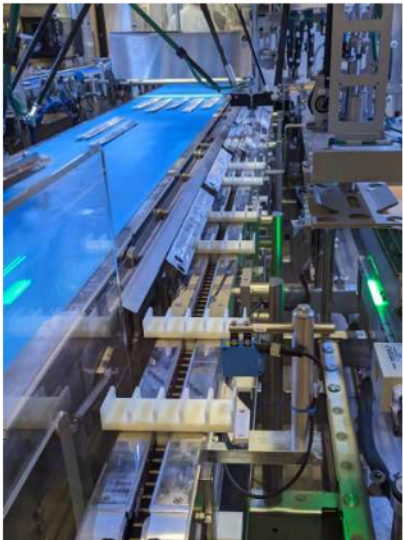
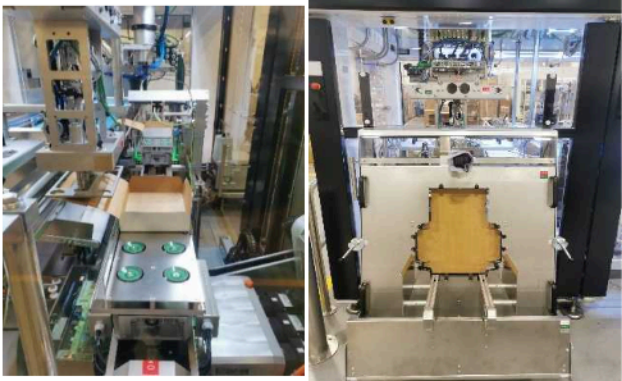
- Takes carton blanks from magazine, applies glue and presses blank into forming box to make carton.
- Carton is then placed on the trans-module.
- With engineering support build empty boxes at the start of each shift, this is to enable operators to package the packs that were not picked by the Robot throughout the shift.
- Operator Packaging consists of using the shoehorn to package 30 pack as per PI-11604. On completion introduce the box to the labelling machine.

Trans-modules

- Carry cartons from one end of the machine to the other along a track
- After finished carton is removed from trans-module, it inverts and travels underneath the track and returns to the carton building station again.

Pack Picking (F4 Robots/Spider robots)

- Two spider robots work in synchronisation to pick packs off the belt and create a buffer, there are 5 packs in each buffer.
- The spider robot picks 5 packs and places them into the carton in layers of 5 until it has reached the desired amount of 30 packs using the tip to tip format. Please see PI- 11578.
- Packs that are overlapping will run through to the rework bin.
- These packs will need to be fed back into the machine using the manual infeed system.



Number (version)	75CLR00674_Att1 (3.0)
Title	Next Generation Pack 2 (NGP2) and Secondary Packaging Machine 2 (SP2) General Overview

MACHINE OVERVIEW

Carton closing/Literature picking

- Robot picks literature from the literature track and places them into the already full carton.
- Robot then folds the flap and closes the lid, holds it to allow the glue to set and then places it onto the outfeed conveyor to be packed into the shipper.

Outfeed conveyor

- Outfeed conveyor carries completed cartons from the F202 robot all the way to the good carton buffer belt.
- Along the way, the carton will pass by reject station 1, in the case that a problem is detected by the machine with the count/orientation of packs etc. during the packing of the carton, the carton will be diverted into reject station 1 using an air blast. Good cartons will continue and turn through 90 degrees and go to the print and apply station.
- If a problem is detected with the print and apply station or the camera does not pass the applied label, the carton will be rejected and reject station 2 (diverted down reject buffer conveyor via air blast).
- Good cartons will continue to the end of the conveyor and be diverted down the good carton buffer conveyor via air blast.
- All 3 buffer conveyors are fitted with sensors to alert the operator to the fact the buffer is full.

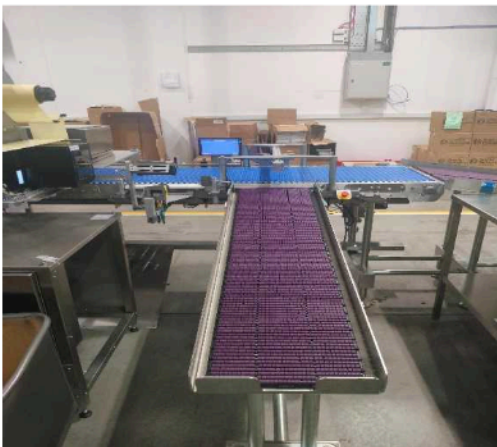
Cartons from both reject stations need to be removed and the packs re-worked using the manual infeed system.



Literature loading



Good carton belt



Reject carton belt 2

Number (version)	75CLR00674_Att1 (3.0)
Title	Next Generation Pack 2 (NGP2) and Secondary Packaging Machine 2 (SP2) General Overview

MACHINE OVERVIEW

Removing jammed cartons from the machine

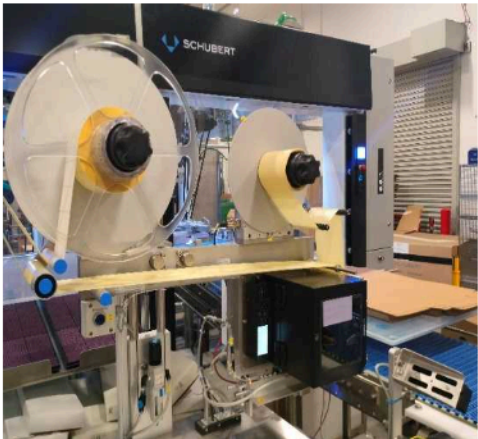
- In the event of a carton jamming on the Trans modules, the machine will stop, and the operator will need to manually remove the cartons from the machine. Operators must remove the cartons from the Trans modules by opening the doors at the back of the machine. Do not try to remove cartons by reaching across the conveyor belt.

Print and apply

- The print and apply station automatically print and applies the control label and country specific label to the carton.
- The leading edge of each carton is detected by an optical sensor, the labeller applies the printed control label to the side of the carton and the country specific to the top of the carton.
- The camera reads the critical information on the label and sends a pass/fail signal to the machine. Reject cartons are rejected at reject station.

Manual infeed

- Catheters from rejected cartons as well as catheters not picked from the spider robots are to be taken to the manual infeed station. This station can only be used when the NGP2 machine is not running, there is a button on the HMI to stop NGP2 outfeed from running.
- This station allows packs to be fed manually into the machine by the operator, place catheters one by one on the belt and they will be fed back into the machine and picked by the spider robots and placed onto the buffer.



Country specific label



Control label



Manual infeed belt

Number (version)	75CLR00674_Att1 (3.0)
Title	Next Generation Pack 2 (NGP2) and Secondary Packaging Machine 2 (SP2) General Overview

HMI OVERVIEW

SP2 has 1 HMI screen at auto infeed conveyer belt.

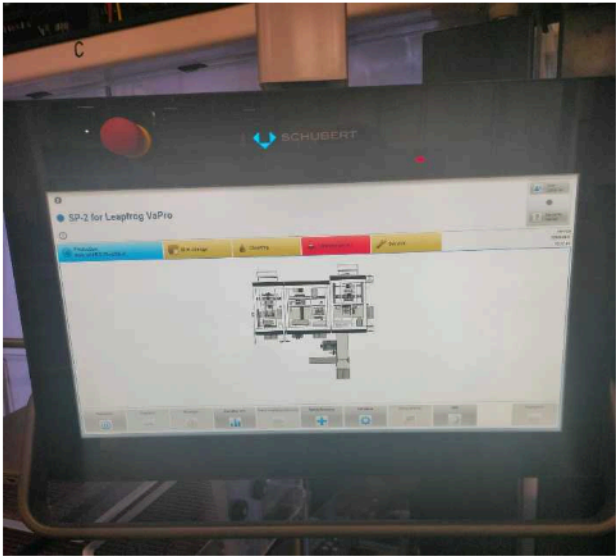
- 1) **Emergency Stop Button** – This button stops the entire machine in the event of danger. The entire machine stops immediately.
- 2) **Stop Button** – Button acknowledges information or error messages, by pressing the reset button error messages are acknowledged
- 3) **Start Button** – Button starts the machine and flashes in the start and stop phase.
- 4) **Touchscreen** – The HMI is used for operating and monitoring the machine.

SP2 has control panels at each door when accessing the machine. There is an emergency stop button, reset button and Start button on each of the panels.

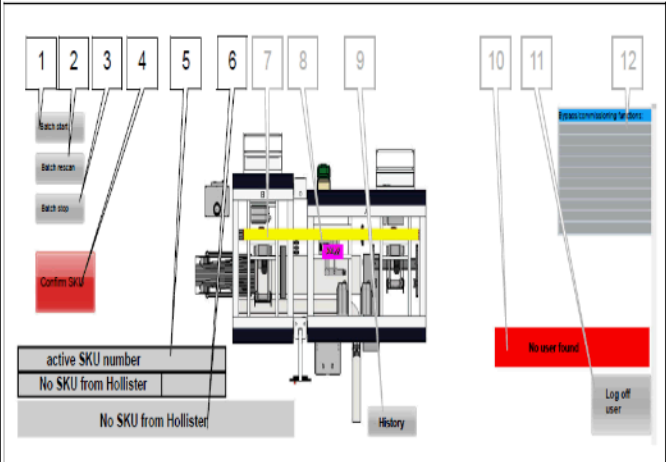
 A Blue reset button will appear on the panels when the doors have been opened.



- 1. The **Start Batch** button starts a new batch. The new batch must be scanned in with the hand scanner
 - 2. The **Batch rescan** button stops the current batch and starts a new one to scan it in with the hand scanner
 - 3. The **Batch stop** button ends the current batch
 - 4. The **Confirm SKU** button confirms the SKU number provided by the operator
 - 5. The **active SKU number** display shows the active registration number provided by the operator
- The display shows the registration number currently scanned in.



SP2 HMI



Number (version)	75CLR00674_Att1 (3.0)
Title	Next Generation Pack 2 (NGP2) and Secondary Packaging Machine 2 (SP2) General Overview

END OF SHIFT

- At the end of each shift, the operator must fill in the relevant details to complete the production/process/parameter sheets and place them on the Group Leader’s desk.
- At the end of the shift, the operator must ensure to report the product count and the amount of calculated waste from the screen into the Handheld Terminal (HHT) (See how to report goods in the SAP – Training Guide for Production Operators 75GEN00060_Att6). The operator must also ensure to complete the relevant parameter checks on the parameter check sheet and place it on the Group Leader’s desk.
- The machine must be always kept neat and tidy.

Note: All surfaces of the Next Generation Pack Machine and associated work benches must be cleaned with 70% IPA twice per shift as per 75CLR00001. A record of this cleaning must be recorded in the Production Sheet 75CLR00674_Att4

To ensure that cleaning is carried out in the correct manner please follow the guidelines below;

- always wear gloves to protect your skin before using IPA
- always spray IPA onto the wipe, not the surface being cleaned - this will prevent IPA being sprayed into the eyes
- always ensure that you use enough IPA to wet the surface being cleaned

Number (version)	75CLR00674_Att1 (3.0)
Title	Next Generation Pack 2 (NGP2) and Secondary Packaging Machine 2 (SP2) General Overview

CHANGEOVERS

NGP2:

When a changeover has occurred on NGP2, the operator must ensure that the area is cleared of all components and the assemblies not required are removed to the racks or the holding areas.

When changing from one SKU to another the operator must click empty run on the HMI, this will empty both modules of catheters and runs only foil in the PMK, select the new SKU recipe which will dump the present packs at St.73. Operators need to remove the catheters from the infeed vibratory track, manually remove all packs on the machine that are after ST.73, the operator must complete a physical check to ensure no other product from current SKU is remaining in the machine.

Confirm against the work order that the correct Labels and Foil are loaded into the machine.

To prevent cross contamination, a line clearance must be performed prior to a changeover.

The Line Clearance for Next Generation Pack (NGP2) consists of:

- 1. Operator must check each pack for LOT number at Start Up.
- 2. Checking each compartment/station for rogue catheters, correct labels etc.
- 3. Check the load stations for rogue catheters, assemblies etc.
- 4. Operator must check each pack for correct Print SKU at Start Up.

A record of the Line Clearance must be recorded in the production sheet 75CLR00674_Att4 for Next Generation Pack Machine

Note: When a work order has been finished, the operator must ensure that Start Up line Clearance, Verification Check and End of Procedure sections are filled out correctly on the work order sheet before placing it on the Team Leaders (TL) desk.

Note: All materials that are removed from the machine must have the original WIP (Work In Progress) ticket reapplied. If the original WIP ticket is not available inform the CTL or GL immediately as they can reprint one from the system. If a material is removed at any other time it must have the original WIP ticket reapplied.

Number (version)	75CLR00674_Att1 (3.0)
Title	Next Generation Pack 2 (NGP2) and Secondary Packaging Machine 2 (SP2) General Overview

CHANGEOVERS

SP2:

The machine has an “Empty” mode to remove all components from the infeed conveyer throughout to the outfeed conveyer.

Prior to changeover the machine should be emptied using the “Empty” function.

When a changeover has occurred, the operator must ensure that the area is cleared of all components and the assemblies not required and are removed to the racks or the holding areas.

When changing from one SKU to another the operator need to only remove the catheters from the feed track and to complete the run out on the machine. The operator must complete a physical check to ensure no cases or catheters remain on the machine indexer, clear out both reject bins and remove cartons from reject conveyer.

To prevent cross contamination, a line clearance must be performed prior to a changeover.

The Line Clearance for Secondary Packaging Machine consists of:

- 1. Checking each compartment/station for rogue catheters, literature, cartons, correct labels etc.
- 2. Check the load stations / bottom of machine for rogue catheters, assemblies etc.